$x(t) = \cos(2\pi t)$

$F_s = 5\,Hz$ olsun

$x[n] = x(t)\Big|_{t=nT} = \cos\frac{2\pi n}{5} = \cos 0{,}4\pi n$

$w = 0{,}4\pi$

$2\pi k = 0{,}4\pi N$

$N = 5k \qquad N_0 = 5$ periyottur.

$x_1[n] = \cos(0{,}4\pi n)$ ← bu işaretin frekansı daha yüksek

$x_2[n] = \cos(0{,}2\pi n)$

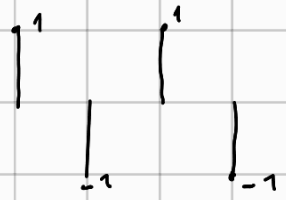
> Daha sık örnekleme aynı işaret için ayrık zamanlı işaretin frekansını azaltır

Ayrık zamanda daha yüksek değişimler daha yüksek frekans demektir

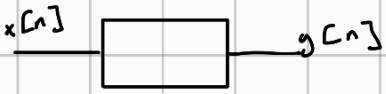
az değişler daha az frekans demektir.

Ayrık zamanda en yüksek frekans $w = \pi$ 'dir

$x[n] = \cos\pi n$



Sgstems

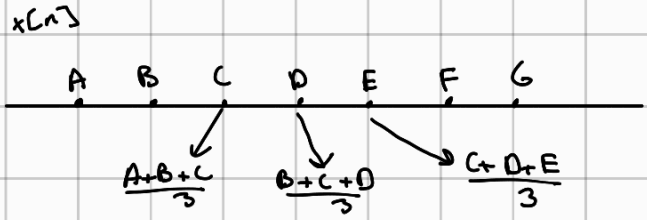


$y[n] = x[n - n_d]$ ⇔ gecikme

bellekli ↙

$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$

ortalama alıyor hareketli
moving average
(son üç örneklemin ortalaması alıyor)

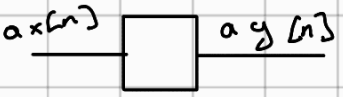


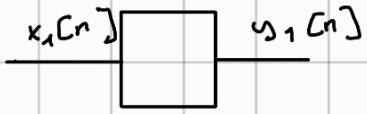
(Alçak geçiren filtre aslında aynı zamanda)

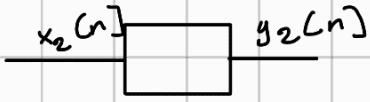
$y[n] = \frac{1}{4} x[n]$ belleksiz

Linear system

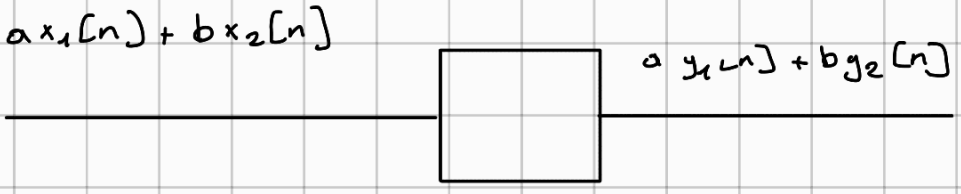
linear system











$y[n] = 8x[n] + 1$

linear sisteme 0 verince e 0 çıkmalıdır

$x_1[n] = 2$ $\quad y_1[n] = 17$

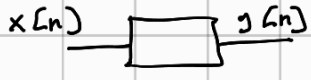
$x_2[n] = 3$ $\quad y_2[n] = 25$

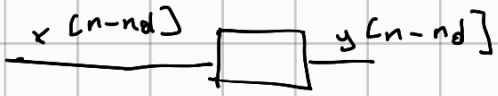
$x_3[n] = 5$ $\quad y_3[n] = 41 \neq 42$

Time invariant Systems

$x_1[n] = y_1[n]$ time invariant ise

$x_1[n-nd] = y_2[n-nd]$





Nedensel sistemler

çıkış şimdiki ve önceki örneklere bağlı

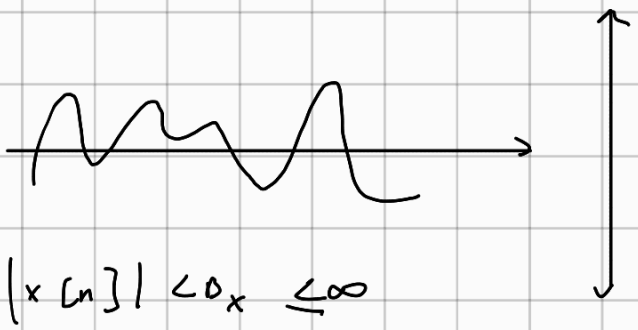
zamansa nedensel olmayan sistemler gerçekleştirilemez

Ama konumsa gerçekleşebilir!!

non-causal - $y[n] = x[n-1] + x[n] + x[n+1]$

anticausal - $y[n] = x[n+1]$ sadece geleceğe bağlı

kararlılık (BIBO)



$|x[n]| < B_x \leq \infty$

$|y[n]| < B_y \leq \infty$

Bir sistemin kararlı olduğunu nasıl anlarız

kararlı olduğunu göstermek çok zor

$y[n] = (x[n])^3$

$|x[n]| < B_x$

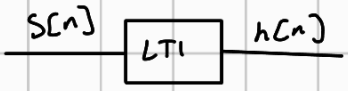
$|y[n]| < (B_x)^3$ → hala sonlu

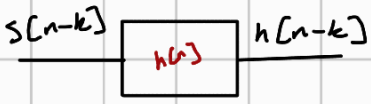
$$y[n] = \begin{cases} 0, & n < 0 \\ n+1 & n \geq 0 \end{cases}$$

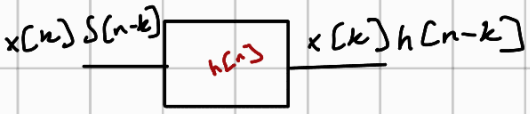
↓ n sonsuza giderken

sistem kararsızdır.

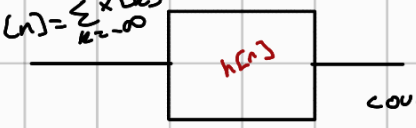
LTI systems





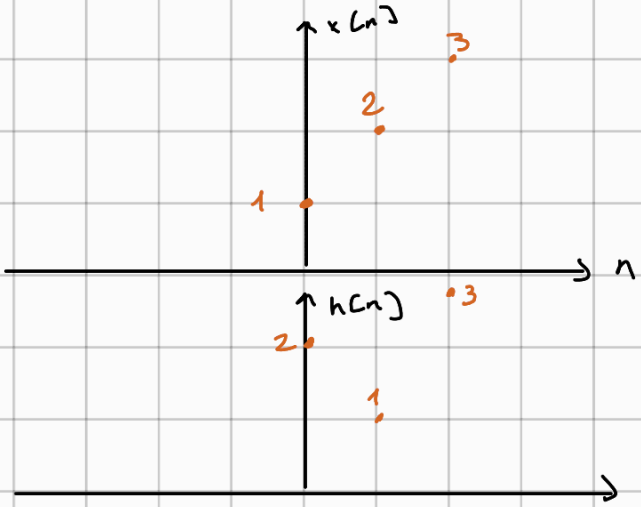


$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$



$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = x[n] * h[n]$

convolution sum

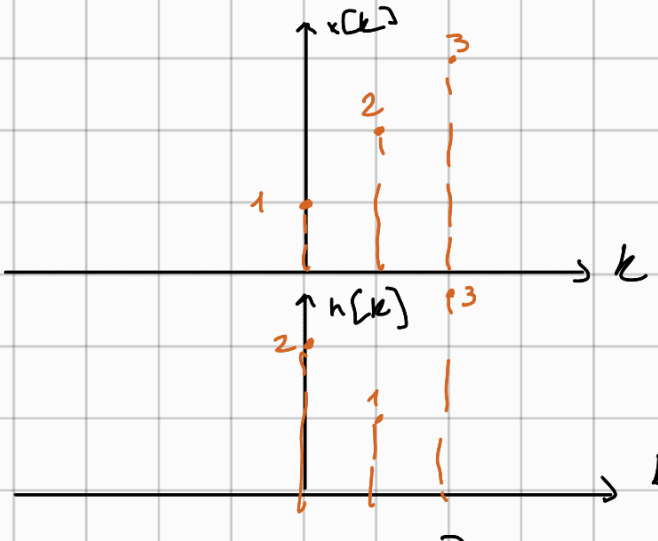


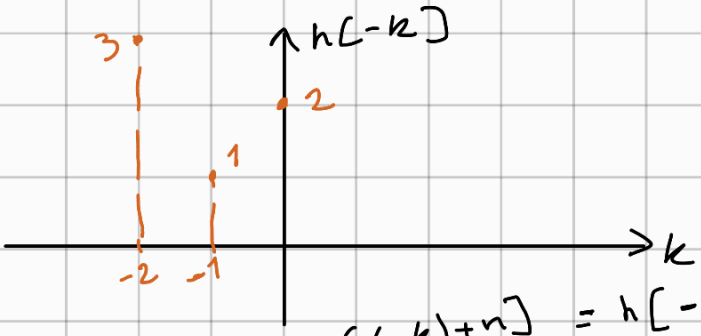
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

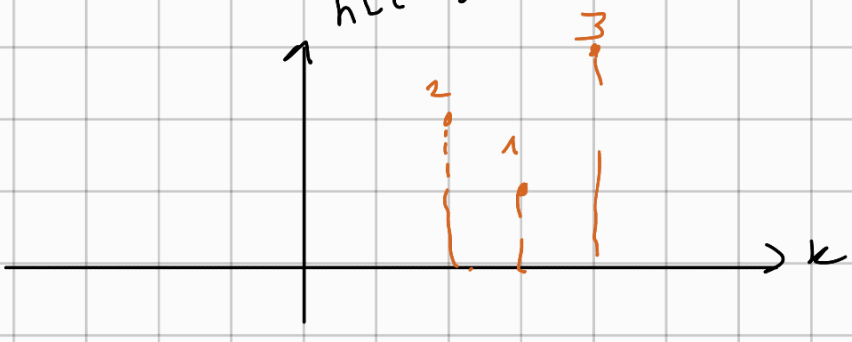
$x[n] = [1\ 2\ 3]$

$h[n] = [2\ 1\ 3]$

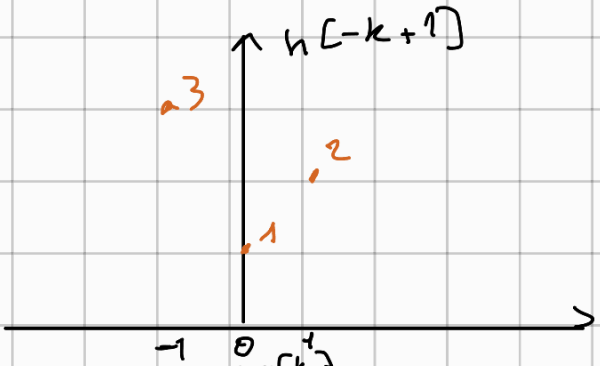




$h[(-k)+n] = h[-(k-n)]$



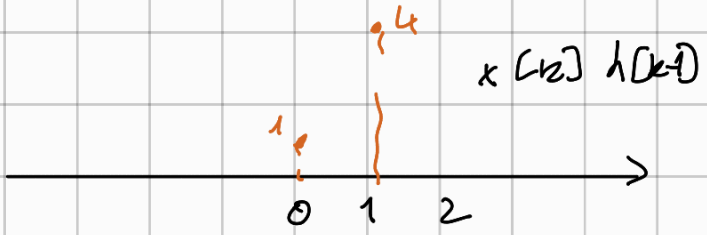
$y[0] = 2.1 = 2$





$y[1] = 1.1 + 2.2 = 5$

$k = 0 \qquad 1$



x boyutu 3 = A

h boyutu 3 = B

y boyutu 5 = A+B-1

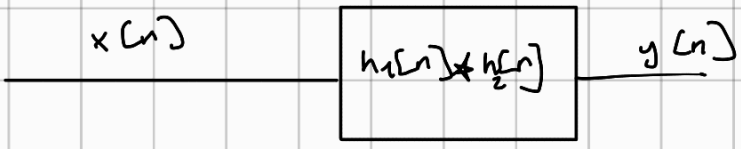


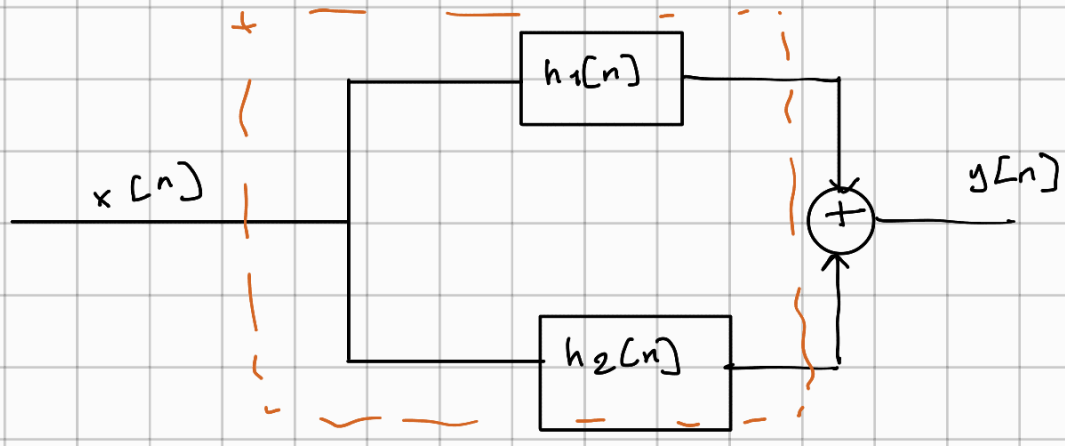
$$w[n] = x[n] * h[n]$$

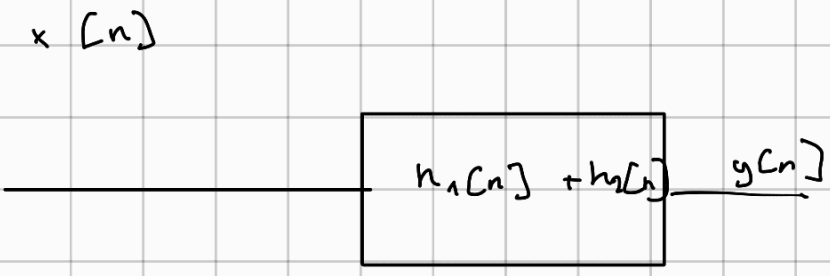
$$y[n] = w[n] * h_2[n]$$

$$= (x[n] * h_1[n]) * h_2[n]$$

$$= x[n] * (h_1[n] * h_2[n])$$







h[n]

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[n] = \cdots + h[-2]\,x[n-2] + h[-1]\,x[n-1] + h[0]\,x[n] + h[1]\,x[n-1] + h[2]\,x[n-2] + \cdots$$

Nedensel sistemlerde çıkış gelecekteki ~~çıkış~~ girişlerden bağımsız

$h[-1] = 0$

$h[-2] = 0$

$\vdots$

$h[-n] = 0$

$h[n] = 0 \quad n < 0$

$\Rightarrow$ sistem nedenseldir.

Kararlılık = Genliği sınırlı bir giriş için çıkışı sınırlı olan sistem

$$\sum_{-\infty}^{\infty} |h[n]| < \infty$$ (mutlak toplanabilir)

Sakın sınavda mutlak koymayı unutma Ori alırsın :)

$h[n] = u[n]$ kararsız

$h[n] = \alpha^n u[n]$

geometrik seri toplamı

toplamı $\frac{1}{1-|\alpha|}$ $|\alpha| < 1$

$|\alpha| \geq 1$ ise kararsızdır sistem

$h[n] = 0,2^n u[n]$ toplamı $\frac{1}{1-|\alpha|} = \frac{1}{1-0,2} = 1,25$

$= 1 + 0,2 + 0,04 + \cdots$

FIR ve IIR

Finite Impulse Response

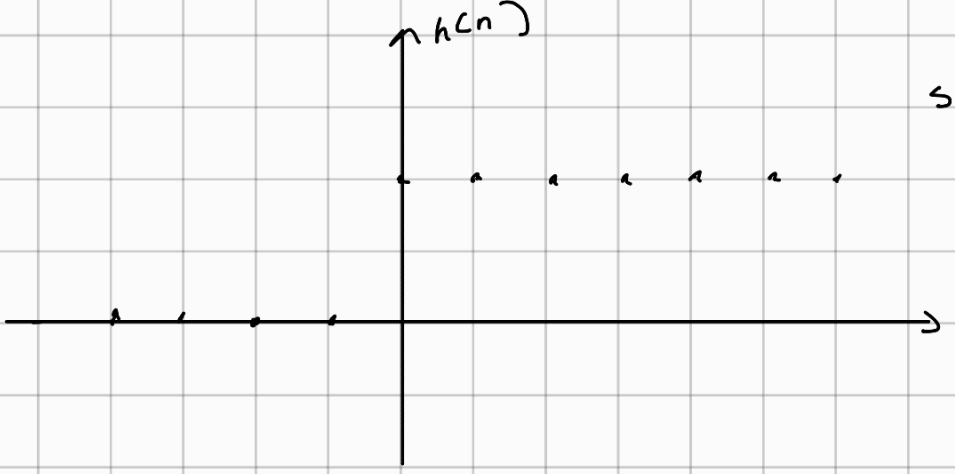
Infinite Impulse Response



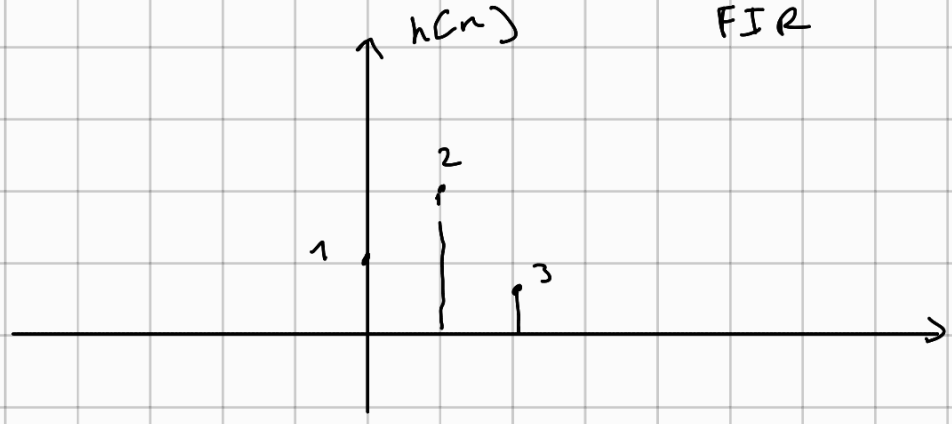
IIR

sonsuz noktada 0'dan farklı değer alır

IIR



sonsuz noktada 0'dan farklı değer alır



FIR

sonlu sayıda 0'dan farklı değer alır

h[n]'in genliği her zaman sonludur

FIR her zaman kararlı

IIR (her zaman) kararlı değil ama kullanırız çünkü daha az pencere (örnek) ile FIR'in yaptığı iş yapılabilir (bazı işaretler için)